

Day-78-02 Azure Virtual Network + NSG (Network Security Group)

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and the user profile 'lovelysh2604@gmail.com'. The main content area shows the 'vnet-demo-lab-1775723896316 | Overview' page. A green checkmark indicates 'Your deployment is complete'. The deployment details show the name 'vnet-demo-lab-1775723896316', subscription 'Azure for Students', and resource group 'rg-demo1'. The start time is '4/9/2026, 2:08:23 PM' and the correlation ID is '9418d0c8-217d-4e7c-82d5-a3fb977699a2'. A sidebar on the right contains links for 'Cost management', 'Microsoft Defender for Cloud', and 'Free Microsoft tutorials'. Below the deployment overview, the 'Subnets' page is shown for the 'vnet-demo-lab' virtual network. A notification 'Successfully added subnet' is visible. The subnets table lists two subnets: 'web-subnet' with IPv4 address space '10.0.1.0/29' and 'app-subnet' with IPv4 address space '10.0.2.0/29'. The bottom of the screen shows the Windows taskbar with the time '2:13 PM'.

Microsoft Azure

Home

vnet-demo-lab-1775723896316 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : vnet-demo-lab-1775723896316

Subscription : Azure for Students

Resource group : rg-demo1

Start time : 4/9/2026, 2:08:23 PM

Correlation ID : 9418d0c8-217d-4e7c-82d5-a3fb977699a2

> Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Find an Azure expert >

Microsoft Azure

Home > vnet-demo-lab-1775723896316 | Overview > vnet-demo-lab

vnet-demo-lab | Subnets

Virtual network

Search

+ Subnet Refresh Manage users Delete Export to CSV

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	web-subnet	10.0.1.0/29	-	3	-	-	-		
☐	app-subnet	10.0.2.0/29	-	3	-	-	-		

Showing 2 subnets

Give feedback

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lovelysh2604@gmail.com
DEFAULT DIRECTORY (LOVELYSH...)

Home > Resource Manager | Resource groups > rg-demo1

vnet-demo-lab
Virtual network

Retrieve detailed routing information for troubleshooting | Review flow metrics for my Virtual Network | How can I troubleshoot virtual network connectivity issues?

Search

Move | Delete | Refresh | Give feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Address space
 - Connected devices
 - Subnets
 - Bastion
 - DDoS protection
 - Firewall
 - Microsoft Defender for Cloud

Essentials

Resource group (move) : rg-demo1

Location (move) : Central India

Subscription (move) : Azure for Students

Subscription ID : 907eda19-180e-44b4-ba2d-b82d41f8f95e

Address space : 10.0.0/16

Subnets : 2 subnets

DNS servers : Azure provided DNS service

BGP community string : Configure

Virtual network ID : cec4393c-3cdf-453d-b9d2-689ae1412f83

Tags (edit) : Add tags

JSON View

Topology | Properties | Capabilities | Recommendations | Tutorials

DDoS protection

Configure additional protection from distributed denial of service attacks.

Not configured

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

Not configured

Peerings

Seamlessly connect two or more virtual networks.

Not configured

Add or remove favorites by pressing CTRL+SHIFT+F

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lovelysh2604@gmail.com
DEFAULT DIRECTORY (LOVELYSH...)

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a VM optimized for high availability | Help me choose the right VM size for my workload | Help me create a low cost VM

Validation passed

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Price

1 X Standard B2ats v2

by Microsoft

Terms of use | Privacy policy

Subscription credits apply ⓘ

0.0055 USD/hr

Pricing for other VM sizes

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

Lovely Sharma

Preferred e-mail address

lovelysh2604@gmail.com

< Previous | Next > | Create

Download a template for automation | Give feedback

Microsoft Azure

Search resources, services, and docs (G+J)

Copilot

lovelysh2604@gmail.com
DEFAULT DIRECTORY (LOVELYSH...)

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260409141600 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2... Start time : 4/9/2026, 2:32:50 PM
Subscription : Azure for Students Correlation ID : 8137a3b7-cb86-4368-9c7a-f4967bef2a02
Resource group : rg-demo1

Deployment details

Next steps

Set up auto-shutdown Recommended
Monitor VM health, performance, and network dependencies Recommended
Run a script inside the virtual machine Recommended

[Go to resource](#) [Create another VM](#) [Scale out your VM](#)

Give feedback

Tell us about your experience with deployment

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Add or remove favorites by pressing CTRL+Shift+F

Search

26°C

ENG 2:34 PM

Microsoft Azure

Search resources, services, and docs (G+J)

Copilot

lovelysh2604@gmail.com
DEFAULT DIRECTORY (LOVELYSH...)

Home > Compute infrastructure | Virtual machines

Compute infrastructure | Virtual mac...

Microsoft

Search

Virtual machines Get started

Overview

All resources

Infrastructure

Virtual machines

Virtual Machine Scale Set (VMSS)

Compute Fleet

Arc Machines

Disks + images

Capacity + placement

Related services

Monitoring + Operations

Help

Showing 1 - 1 of 1. Display count: auto

vm-web

Virtual machine

Search

Help me copy this VM in any region

Manage this VM with Azure CLI

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Essentials

Resource group (move) : rg-demo1

Status : Running

Location : Central India (Zone 2)

Subscription (move) : Azure for Students

Subscription ID : 907eda19-180e-44b4-ba2d-b82d...

Availability zone : 2

Operating system : Linux (ubuntu 24.04)

Size : Standard B2ats v2 (2 vcpus, 1 GiB ...)

Primary NIC public IP : 20.197.1.198
1 associated public IPs

Virtual network/subnet : vnet-demo-lab/web-subnet

DNS name : Not configured

Health state : -

Time created : 4/9/2026, 9:02 AM UTC

JSON View

Add or remove favorites by pressing CTRL+Shift+F

Search

26°C

ENG 2:39 PM

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lovelysh2604@gmail.com
DEFAULT DIRECTORY: LOVELYSH...

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Validation passed

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Price

1 X Standard DC1ds v3
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ
0.0222 USD/hr
[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

Lovely Sharma

Preferred email address

lovelysh2604@gmail.com

< Previous

Next >

Create

[Download a template for automation](#) [Give feedback](#)

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lovelysh2604@gmail.com
DEFAULT DIRECTORY: LOVELYSH...

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260409144243 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2...

Start time : 4/9/2026, 2:47:53 PM

Subscription : Azure for Students

Correlation ID : d1854a62-c8dd-4a7a-8071-80ec152870c9

Resource group : rg-demo1

> Deployment details

✓ Next steps

Set up auto-shutdown Recommended

Monitor VM health, performance, and network dependencies Recommended

Run a script inside the virtual machine Recommended

Go to resource

Create another VM

Scale out your VM

Give feedback

[Tell us about your experience with deployment](#)

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

[Set up cost alerts >](#)

Add or remove favorites by pressing Ctrl+Shift+F

portal.azure.com/view/Microsoft_Azure_ComputeHub/ComputeHubMenuBlade/~virtualMachinesBrowse

Microsoft Azure

Home > Compute infrastructure

Compute infrastructure | Virtual machines

Clone a selected virtual machine | List all alerts for virtual machines | Recommend VM sizes for my workloads

Search

Virtual machines | Get started

Overview

All resources

Infrastructure

Virtual machines

Virtual Machine Scale Set (VMSS)

Compute Fleet

Arc Machines

Disks + images

Capacity + placement

Related services

Monitoring+Operations

Help

Create | Reservations | Manage view | Refresh | Export to CSV | Open query | Assign tags | Start | Restart | ... | Group by none

Filter for any field...

Subscription equals all | Type equals all | Resource Group equals all | Location equals all | Add filter

Name ↑	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
vm-app	Azure for Stude...	rg-demo1	Central India	Running	Linux	Standard_DC1d...	20.193.159.216	1
vm-web	Azure for Stude...	rg-demo1	Central India	Running	Linux	Standard_B2ats...	20.197.1.198	1

Showing 1 - 2 of 2. Display count: auto

Give feedback

Add or remove favorites by pressing Ctrl + Shift + F

Microsoft Azure

Home > Compute infrastructure | Virtual machines > vm-web

Compute infrastructure | Virtual mac...

vm-web | Connect

Virtual machine

Connection prerequisites

Switch to PowerShell | Restart | Manage files | New session | Editor | Web preview | Settings | Help

Requesting a Cloud Shell. Succeeded.

Connecting terminal...

```
lovely [ ~ ]$ ssh Lovely-web@20.197.1.198
The authenticity of host '20.197.1.198 (20.197.1.198)' can't be established.
ED25519 key fingerprint is SHA256:RguLKabExB+QgPLyVUYHjZdK2zqTEOuUFGNKMk4bzh0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.197.1.198' (ED25519) to the list of known hosts.
Lovely-web@20.197.1.198's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Apr  9 09:31:01 UTC 2026

System load:  0.08      Processes:      131
Usage of /:   5.7% of 28.02GB   Users logged in:  0
Memory usage: 33%      IPv4 address for eth0: 10.0.1.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
```

```
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
lovelysh2604@gmail.com
DEFAULT DIRECTORY (LOVELYSH...)

Home > Compute infrastructure | Virtual machines > vm-web

Compute infrastructure | Virtual mac...
vm-web | Connect
Connection prerequisites

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Lovely-web@vm-web:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1881 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [343 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1663 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [322 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2911 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [674 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:24 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:25 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7520 B]
```

```
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
lovelysh2604@gmail.com
DEFAULT DIRECTORY (LOVELYSH...)

Home > Compute infrastructure | Virtual machines > vm-web

Compute infrastructure | Virtual mac...
vm-web | Connect
Connection prerequisites

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Lovely-web@vm-web:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 32 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 0s (12.7 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 69217 files and directories currently installed.)
Preparing to unpack .../nginx-common-1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx-1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
* Upgrading binary nginx
[ OK ]
```

Microsoft Azure

Home > Compute infrastructure | Virtual machines > vm-web

Compute infrastructure | Virtual mac... | vm-web | Network settings

Virtual machines

- Overview
- All resources
- Infrastructure
 - Virtual machines
 - Virtual Machine Scale Set (VMSS)
 - Compute Fleet
 - Arc Machines
 - Disks + images
 - Capacity + placement

Showing 1 - 2 of 2. Display count: auto

vm-web | Network settings

- Networking
 - Network settings
 - Load balancing
 - Application security groups
 - Network manager
- Settings
 - Availability + scale
 - Security
 - Backup + disaster recovery

Add inbound security rule

vm-web-nsg

Source: Any

Source port ranges: *

Destination: Any

Service: Custom

Destination port ranges: 80

Add Cancel Give feedback

2489 "nginx: master process /usr/sbin/nginx -p daemon on; master process on;"

